FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
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ALL INDIA TEST SERIES

FIITJEE

# JEE(Advanced)-2015

## ANSWERS, HINTS & SOLUTIONS
## FULL TEST– IV
## (Paper-1)

| Q. No. | PHYSICS | CHEMISTRY | MATHEMATICS |
|---|---|---|---|
| 1. | B | A | C |
| 2. | C | C | B |
| 3. | A | D | B |
| 4. | D | A | A |
| 5. | C | A | C |
| 6. | B | A | D |
| 7. | B, D | A, D | B, C |
| 8. | A, B, C, D | A, B, D | B, D |
| 9. | C, D | A, B, C | B, C |
| 10. | A | A | C |
| 11. | C | B | C |
| 12. | C | C | C |
| 13. | B | B | B |
| 14. | A | A | A |
| 1. | A → p<br>B → r<br>C → s<br>D → q | A → p, r, s, t<br>B → p, r<br>C → q<br>D → p, r, s | (A) → (r)<br>(B) → (s)<br>(C) → (p, r)<br>(D) → (p, q, r, s, t) |
| 2. | A → s<br>B → r<br>C → q<br>D → q | A → s<br>B → p<br>C → q<br>D → q | (A) → (r)<br>(B) → (t)<br>(C) → (q)<br>(D) → (s) |
| 1. | 3 | 3 | 4 |
| 2. | 5 | 3 | 6 |
| 3. | 5 | 4 | 6 |
| 4. | 5 | 6 | 2 |
| 5. | 8 | 7 | 2 |

# Physics

## PART – I

## SECTION – A

1. If the pulley comes down by a displacement x then the block comes down by 3x and the string connecting the block has tension Kx.

$\Rightarrow -Kx = M \times \frac{3d^2x}{dt^2}$

or $\omega = \sqrt{\frac{K}{3M}}$.

2. $\frac{1}{v} + \frac{1}{u} = \frac{1}{F_{eff}}$

$\Rightarrow \frac{1}{20/3} + \frac{1}{-20} = \frac{1}{F_{eff}}$ or $F_{eff}$ = +10 cm

Now $\frac{1}{F_{eff}} = \frac{1}{f_m} - \frac{2}{f_0}$

or $\frac{1}{10} = \frac{1}{f_m} + \frac{2}{30}$

or $\frac{1}{f_m} = \frac{1}{10} - \frac{2}{30} = \frac{1}{30}$

or $f_m$ = 30 cm or R = +60 cm.

3. Use KCL/KVL or suppose the current due to each battery. (Forms wheatstones by removing 1 battery at a time).

4. There is a tetrahedral configuration with a total of 6 interactions.

5. Say the beam is incident along $\hat{I} = \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$ and the normal of one of the faces is $\hat{N} = \hat{i}$ then its projection area is $A\hat{N}.\hat{I} = \frac{a^2}{\sqrt{3}}$. There are three such faces on which the radiation falls.

$\Rightarrow$ Total projection area = $\sqrt{3}a^2$

$\Rightarrow$ Force = $\frac{I}{c}.\sqrt{3}a^2$.

6. Mechanical energy and momentum both have to be conserved. In case of rough incline, the sphere will also have some rotational KE. Thus lesser energy available as translational KE of incline as well as the cylinder.

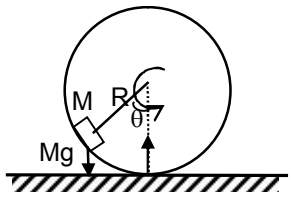
7. Speed is characteristic of the medium is same for all observers. Here wavelength is different for different directions and thus frequency.
For observer (a, 0), (0, a) and (0, –a) observed frequency is more than emitted taking into account time lag.

8. Potential difference between plates is same as $\varepsilon$.
$V_A - V_B = \varepsilon$ = E.d in both air and liquid.
$V_A = V_B$ and $V_A' = V_B'$
Complete plate is at same potential.

9. Since coils carry current in opposite sense magnetic field is always zero in solenoid even if current is there. Thus net inductance = 0, it is simply $\varepsilon$ and R joined in series.

10. Instantaneous axis has zero.
instantaneous acceleration $\omega = 0$
Taking $\tau$ about bottom most point:
$Mg\ R\sin\theta = m(2R\sin\theta/2)^2\ \alpha$.
Solving we get,

$$\alpha = \frac{g\sin\theta}{4R\sin^2\left(\frac{\theta}{2}\right)}$$




11. $mg - N = mR\alpha\sin\theta$
$\eta N = m(g - R\alpha\sin\theta)$

12-14. Use Doppler's effect, the speed of sound increases with increase in temperature.

## SECTION - B

1. $N_A + N_B + N_C = N_0$.
B will first have more production rate than decay and finally there will be no B.

2. In steady state all capacitors G have as open circuits and inductors as short circuits.
$\Rightarrow$ purely resistive network:
Assume, $i = aE_1 + bE_2$
We get, a = 2, b = 3.
$\Rightarrow i = 2E_1 + 3E_2$.

## SECTION – C

1. There is only a single atom, thus maximum number of photons obtained are
$4 \rightarrow 3$, $3 \rightarrow 2$ and $2 \rightarrow 1$.

2. Surface tension force is acting downward at both inner and outer surface at contact point on the tube.
$F = 2\pi rT\cos\theta + 2\pi rT\cos\theta = 4\pi rT\cos\theta$

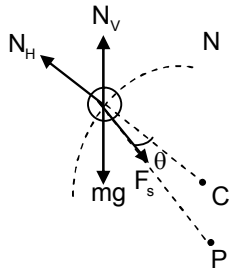
3. $(T_A - T_B) = T_0 e^{-\alpha t}$
Same fractional change takes same time, where $T_0 = 80°C$ is initial temperature difference. Since in fixed one hour temperature loss of A = temperature gain of B, means heat capacities are same, means they will always undergo same loss and same gain. After one hour temperature difference become half = $40°C = \frac{1}{2}(80°)$. In further one hour temperature difference becomes further half.
$T_A + T_B = 120°C$
$T_A - T_B = \left(\frac{1}{2}\right)^2 \times T_0 = 20°C$
$\Rightarrow T_B = 50°C \Rightarrow x = 5$

4. Vertical component of normal reaction balances mg and horizontal component of normal reaction balances component of spring force towards centre.

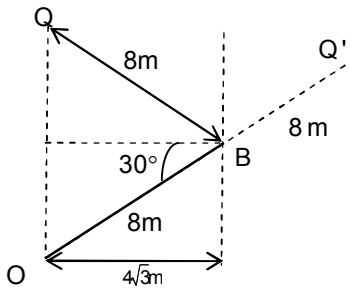
$N_v = mg$

$N_H = F_s \cos\theta = \frac{kR}{4} \times \frac{4}{5} = \frac{k}{5}R$

$N = \sqrt{N_v^2 + N_H^2} = 5$ N.




5. Ball collide at highest point of projectile Motion, after collision, motion is mirror image of motion that would have been without collision.
Top view of motion is as shown in diagram OQ is required distance.
OQ = 2 (OB) sin 30° = 8 m



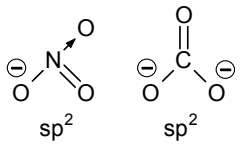
# Chemistry

## PART – II

## SECTION – A

1. Singlet carbene gives *syn* addition on alkene. *syn* addition on cis alkene gives meso compound.

4. $\Delta H = \Delta E + \Delta nRT$, $\Delta H < \Delta E$ if $\Delta n < 0$

5. 


6. The mass of 1 cc of $(C_2H_5)_4Pb$ is = $1 \times 1.66 = 1.66$g and this is the amount needed per litre.

No. of moles of $(C_2H_5)_4Pb$ needed = $\frac{1.66}{323}$ = 0.00514 ml

1 mole of $(C_2H_5)_4Pb$ requires $4 \times (0.00514) = 0.0206$ ml of $C_2H_5Cl$

$\therefore$ Mass of $C_2H_5$ Cl = $0.0206 \times 64.5 = 1.33$ g

8. When a solid mixture of NaCl, $K_2Cr_2O_7$ and conc. $H_2SO_4$ is heated, the products obtained are

$$4KCl + K_2Cr_2O_7 + 6H_2SO_4 \longrightarrow \underset{\text{Red vapour}}{2CrO_2Cl_2} \uparrow + 6KHSO_4 + 3H_2O$$

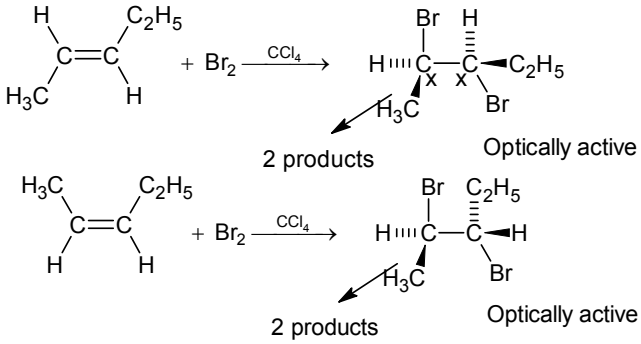
$$CrO_2Cl_2 + 4NaOH \longrightarrow \underset{\text{Yellow solution}}{Na_2CrO_4} + 2NaCl + 2H_2O$$

9. Claisen condensation can be shown only when at least 2 $\alpha$ H are there.

## SECTION-B

2. (A) $K_P > Q$ the reaction will proceed in forward direction spontaneously.
(B) $\Delta G^\circ < RT\log_e Q$ then $\Delta G = +ve$ then non spontaneous.
(C) $K_P = Q \rightarrow$ equilibrium
(D) $\Delta G = \Delta H - T.\Delta S$

## SECTION – C

3. 


total = 2 + 2 = 44. Conductivity of $Na_2SO_4 = 2.6 \times 10^{-4}$

$$\Lambda_m\left(Na_2SO_4\right) = \frac{1000 \times 2.6 \times 10^{-4}}{0.001} = 260 \text{ S cm}^2$$

$$\Lambda_m\left(SO_4^{2-}\right) = \Lambda_m\left(Na_2SO_4\right) - 2\Lambda_m\left(Na^+\right)$$

$= 260 - 2 \times 50 = 160$ S cm$^2$ mol$^{-1}$

Coductivity of $CaSO_4$ solution

$= 7 \times 10^{-4} - 2.6 \times 10^{-4} = 4.4 \times 10^{-4}$ S cm$^{-1}$

$\Lambda_m\left(CaSO_4\right) = \Lambda_m\left(Ca^{2+}\right) + \Lambda_m\left(SO_4^{2-}\right)$

$= 120 + 160 = 280$ S cm$^2$ mol$^{-1}$

Solubility, $S = \dfrac{1000 \times K}{\Lambda_m} = \dfrac{1000 \times 4.4 \times 10^{-4}}{280}$

$= 1.57 \times 10^{-3}$ cm

$K_{sp} = \left[Ca^{2+}\right]\left[SO_4^{2-}\right]_{total} = (0.00157)(0.00157 + 0.001)$

$= 4 \times 10^{-6}\ M^2$

$= y \times 10^{-x}\ M^2$

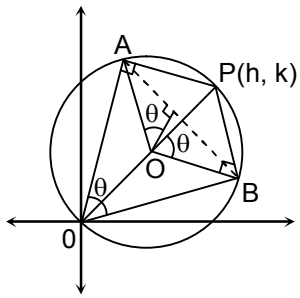
# Mathematics

## PART – III

## SECTION – A

1. Here, $\tan\theta = \sqrt{\frac{3}{2}}$

Thus, locus of P is
$x^2 + y^2 = 180$
Hence, radius = $6\sqrt{5}$




2. Sum = $\text{Re}(e^{i\alpha}(1 + e^{i\alpha})^n)$

3. For the given curve, $\left.\frac{dy}{dx}\right|_{x=3} = \frac{2}{k} = -2 \Rightarrow k = -1$

$\Rightarrow$ The curve is, $x^2 + 2x + 4y + 3 = 0$
and the tangent is $4x + 2y - 3 = 0$

4. Equation of first ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Vertices of second ellipse are $(\pm ae, 0)$

$\Rightarrow$ Equation of second ellipse is $x^2 + \sqrt{2}y^2 = \frac{a^2}{2}$

$\Rightarrow$ Equation of tangent to it at (h, k) is $xh + \sqrt{2}yk = \frac{a^2}{2}$

Homogenising $x^2 + y^2 - b^2 = 0$ with this straight line $x^2 + y^2 - b^2\left[\frac{2}{a^2}\left(xh + \sqrt{2}yk\right)\right]^2 = 0$

These are perpendicular, hence $h^2 + 2x^2 = \frac{a^4}{2b^2} = a^2$

$\Rightarrow$ Locus of (h, k) is $x^2 + 2y^2 = a^2$

5. Given, $\lim\limits_{x\to\infty} \frac{2(x)^{1/2} + 3(x)^{1/3} + 4(x)^{1/4} + ..... + n(x)^{1/n}}{(2x-3)^{1/2} + (2x-3)^{1/3} + ..... + (2x-3)^{1/n}}$

$= \lim\limits_{h\to 0} \frac{2(h)^{-1/2} + 3(h)^{-1/3} + ..... + n(h)^{-1/n}}{h^{-1/2}(2-3h)^{1/2} + h^{-1/3}(2-3h)^{1/3} + ..... + h^{-1/n}(2-3h)^{1/n}}$ [Put $x = \frac{1}{h}$]

$= \lim\limits_{h\to 0} \frac{2 + 3h^{\left(\frac{1}{2}-\frac{1}{3}\right)} + 4h^{\left(\frac{1}{2}-\frac{1}{4}\right)} + ..... + nh^{\left(\frac{1}{2}-\frac{1}{n}\right)}}{(2-3h)^{1/2} + h^{\left(\frac{1}{2}-\frac{1}{3}\right)}(2-3h)^{1/3} + ..... + h^{\left(\frac{1}{2}-\frac{1}{n}\right)}(2-3h)^{1/n}}$

$= \frac{2 + 0 + 0 + ..... + 0}{2^{1/2} + 0 + 0 + .....} = \sqrt{2}$

6. Substitute $x^2 = t$

$\Rightarrow I = \frac{1}{2}\int e^{t^2}\left(1+t+2t^2\right)e^t dt = \frac{1}{2}\int e^t\left[t\,e^{t^2}+\left(e^{t^2}+2t^2\,e^{t^2}\right)\right]dt$

$= \frac{1}{2}\int e^t\left[f(t)+f'(t)\right]dt = \frac{1}{2}e^t\left(te^{t^2}\right)+c = \frac{1}{2}e^{x^2}\left(x^2\,e^{x^4}\right)+c$

7. Let, $f(x) = A + Bx + Cx^2 + Dx^3 + \ldots..$
Now, $f(0) = 0$ and $f'(0) = 0$

8. Here, $f'(a) = \int_0^1 \frac{x^{\cos a}\ln x(-\sin a)}{\ln x}dx = -\int_0^1 x^{\cos a}\sin a\,dx = -\tan\left(\frac{a}{2}\right)$

$\Rightarrow f(a) = -\int\frac{\sin a}{1+\cos a}da = \ln(1+\cos a)$

9. Here, $f'(x) = 3x^2 - 2x + 100 > 0$
$\Rightarrow f(x)$ is increasing

10. Here, $\frac{bx}{c}-\frac{cy}{a}-\frac{az}{b} = b\sin B + c\sin C + a\sin A = \frac{b^2+c^2+a^2}{2R}$
$\Rightarrow k = 2R$

11. $\cot A + \cot B + \cot C = \frac{R}{abc}\left(b^2+c^2-a^2+c^2+a^2-b^2+a^2+b^2-c^2\right)$

$= \frac{R}{abc}\left(a^2+b^2+c^2\right) = \Delta\left(\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}\right)$
$\Rightarrow k = \Delta$

12.-14. Let (h, k) be the circumcircle of $\Delta ABC$
Then, $h = \frac{4+3p}{2p}$; $k = \frac{4-3p}{2q}$
$\Rightarrow p = \frac{4}{2h-3}$; $q = \frac{4h-12}{k(2h-3)}$
This (p, q) lies on $x^2 = 4ay$
$\Rightarrow$ Locus of (h, k) is $\frac{1}{2a}\left(y+\frac{9}{8}a\right) = \left(h-\frac{9}{4}\right)^2$

## SECTION – B

1. (A) A is idempotent, $A^2 = A^3 = A^4 = \ldots.. = A$
$(A + I)^n = I + {}^nC_1A + {}^nC_2A^2 + \ldots.. + {}^nC_nA^n = I + {}^nC_1A + {}^nC_2A + \ldots.. + {}^nC_nA$
$= I + \left({}^nC_1 + {}^nC_2A^2 + \ldots.. + {}^nC_n\right)A = I + (2^n - 1)A$
$\Rightarrow 2^n - 1 = 127 \Rightarrow n = 7$
(B) $(I - A)(I + A + A^2 + \ldots.. + A^7) = (I + A + A^2 + \ldots.. + A^7) - (A + A^2 + \ldots.. A^8)$
$= I - A^8 = I$ (if $A^8 = 0$)
(C) A is skew symmetric $\Rightarrow |A| = |A^T| = (-1)^n|A|$
$\Rightarrow |A|\left(I-(-1)^n\right) = 0$ as n is odd, hence $|A| = O \Rightarrow$ A is singular
(D) A is symmetric then $A^{-1}$ is also symmetric for matrix of any order

2. (A) Required event will occur if last digit in all the numbers is 1, 3, 7 or 9. Thus, required probability = $\frac{4^n}{10^n}$

(B) Required probability = P (last digit is 1, 2, 3, 4, 6, 7, 8, 9) – P (last digit is 1, 3, 7, 9) = $\frac{8^n - 4^n}{10^n}$

(C) P (1, 3, 5, 7, 9) – P (1, 3, 7, 9) = $\frac{5^n - 4^n}{10^n}$

(D) P (required) = P (0, 5) – P (5) = $\frac{\left(10^n - 8^n\right) - \left(5^n - 4^n\right)}{10^n} = \frac{10^n - 8^n - 5^n + 4^n}{10^n}$

## SECTION – C

1. Let hyperbola be $xy = c^2$. Then points of intersection is $\left(ct, \frac{c}{t}\right)$

$\Rightarrow c^2t^4 - 2ct^3 - 20t^2 - 4ct + c^2 = 0$

If $t_1, t_2, t_3$ and $t_4$ are its roots

Then, $\sum t_1 = \frac{2}{c}$; $\sum t_1t_2 = -\frac{20}{c^2}$; $\sum t_1t_2t_3 = \frac{4}{c}$ and $t_1t_2t_3t_4 = 1$

$\Rightarrow \ell = 2, m = 44, n = 56$

2. Let, $y = \left|\sqrt{1+\sin 2x} - \sqrt{1-\sin 2x}\right| \Rightarrow y^2 = 2 - 2|\cos 2x|$

If $x \in \left[0, \frac{\pi}{4}\right]$ or $\left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$ or $\left[\frac{7\pi}{4}, 2\pi\right]$, cos 2x is non-negative

$\Rightarrow y = 2|\sin x| \Rightarrow \cos x \le |\sin x|$ except for x in $\left[0, \frac{\pi}{4}\right]$ and $\left[\frac{7\pi}{4}, 2\pi\right]$

So, it holds for $\left[\frac{3\pi}{4}, \frac{5\pi}{4}\right]$ in which $\sin x \le \frac{1}{\sqrt{2}}$

If $x \in \left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ or $\left(\frac{5\pi}{4}, \frac{7\pi}{4}\right)$ then $\cos 2x < 0 \Rightarrow y = 2|\cos x|$

So, both inequalities hold

Thus, solution set is $x \in \left[\frac{\pi}{4}, \frac{7\pi}{4}\right]$

$\Rightarrow \mu - \lambda = 6$

3. Here, $(1 - 2x + 5x^2 + 10x^3)\left({}^nC_0 + {}^nC_1x + {}^nC_2x^2 + .....\right) = 1 + a_1x + a_2x^2 + .....$

$\Rightarrow a_1 = n - 2$; $a_2 = \frac{n(n-1)}{2} - 2n + 5$

Since, $a_1^2 = 2a_2 \Rightarrow n = 6$

4. $f(x)(1 - x)^{-1} = f(x)(1 + x + x^2 + .....) = b_0 + b_1x + b_2x^2 + .....$

$\Rightarrow (a_0 + a_1x + a_2x^2 + ..... + a_nx^n + .....)(1 + x + x^2 + ..... + x^n + .....) = b_0 + b_1x + b_2x^2 + .....$

Comparing coefficient of $x^n$ on both sides, $a_0 + a_1 + a_2 + ..... + a_{n-1} + a_n = b_n$

Comparing coefficient of $x^{n-1}$ on both sides, $a_0 + a_1 + a_2 + ..... + a_{n-1} = b_{n-1}$

$\Rightarrow b_n - b_{n-1} = a_n$

Also, $b_0 = a_0$ and $b_1 = a_0 + a_1$

$\Rightarrow b_0 = 1; a_1 = 2$

$\Rightarrow b_{10} = a_0 + a_1 + a_2 + ..... + a_{10} = 2^{11} - 1$

5. Here, R.H.S. $= \tan^{-1}\left\{\dfrac{\tan^2(\alpha+\beta)\tan^2(\alpha-\beta)+1}{1-\tan^2(\alpha+\beta)\tan^2(\alpha-\beta)}\right\}$

$= \tan^{-1}\left\{\dfrac{\sin^2(\alpha+\beta)\sin^2(\alpha-\beta)+\cos^2(\alpha+\beta)\cos^2(\alpha-\beta)}{\cos^2(\alpha+\beta)\cos^2(\alpha-\beta)-\sin^2(\alpha+\beta)\sin^2(\alpha-\beta)}\right\}$

$= \tan^{-1}\left\{\dfrac{(\cos 2\beta-\cos 2\alpha)^2+(\cos 2\alpha+\cos 2\beta)^2}{2\cos 2\alpha\cos 2\beta}\right\}$

$= \tan^{-1}\left\{\dfrac{\cos 2\alpha\sec 2\beta+\cos 2\beta\sec 2\alpha}{2}\right\}$

$\Rightarrow \lambda = 2$